

CR-032: Ascending and Descending Order Sorting of the Admin Student List

1. Change Request Details

Field	Details
Change Request ID	CR-032
Title	Ascending and Descending Order Sorting of the Admin Student List
Priority	Medium
Difficulty	Low
Change Type	Frontend Enhancement
Affected Version	Student Management System v1.0 baseline
Dependency	Existing <code>GET /api/students/</code> endpoint
Affected User	Administrator

2. Business Requirement

Currently, administrators can view the complete student list, but there is no sorting facility available. As the number of student records increases, it becomes difficult to quickly locate and organize students.

The system should provide administrators with an easy way to sort student records in **Ascending** and **Descending** order.

The sorting feature will improve the usability of the Admin Dashboard and make student records easier to locate and manage.

3. Current Behaviour

- `DashboardPage.jsx` loads students using `GET /api/students/`.
- The complete student list is displayed in a table.
- Student records are currently displayed in the order returned by the API.
- No sorting facility is currently available.
- Administrators must manually scan the table to locate records.
- Existing Edit and Delete operations use `student_id`.
- Existing administrator authentication and authorization are already implemented.

4. Requested Change

Add an Ascending and Descending sorting facility to the administrator's student table.

The implementation must:

- Provide an **Ascending** sorting option.
- Provide a **Descending** sorting option.
- Allow sorting based on relevant student fields.
- Support sorting by **Registration Number**.
- Support sorting by **Student Name**.
- Perform sorting on already-loaded student records.
- Avoid unnecessary API requests during sorting.
- Display the sorted records immediately.
- Preserve existing Edit and Delete functionality.
- Ensure Edit/Delete continue to use the correct `student_id`.
- Maintain existing administrator authentication and authorization.

5. Scope

In Scope

- Admin Dashboard sorting control.
- Ascending order sorting.
- Descending order sorting.
- Sorting by Registration Number.
- Sorting by Student Name.
- Client-side sorting.
- Sorting already-loaded student records.
- Responsive sorting interface.
- Preservation of existing CRUD functionality.

Out of Scope

- Database changes.
- Backend changes.
- New API endpoints.
- Server-side sorting.
- Pagination.
- Changes to authentication or authorization.
- Changes to existing CRUD operations.

6. Expected / Potentially Affected Files

- `frontend/src/pages/DashboardPage.jsx`
- `frontend/src/components/StudentSortControl.jsx` (*optional*)
- Existing dashboard stylesheet

Students may modify additional files if required. Any additional files must be identified and justified in the Implementation Report.

7. Functional & Security Requirements

Functional Requirements

FR-001: Sorting controls shall be available on the Administrator Dashboard.

FR-002: Administrators shall be able to sort student records in Ascending order.

FR-003: Administrators shall be able to sort student records in Descending order.

FR-004: Sorting shall support Registration Number.

FR-005: Sorting shall support Student Name.

FR-006: Sorting shall be performed against already-loaded student records.

FR-007: Sorting shall not require a new API request for every sorting action.

FR-008: The student table shall immediately reflect the selected sorting order.

FR-009: Existing Edit and Delete functionality shall continue to work correctly.

FR-010: Edit and Delete operations shall continue to use the correct `student_id`.

Security Requirements

SEC-001: Existing administrator authentication shall remain unchanged.

SEC-002: Existing administrator authorization shall remain unchanged.

SEC-003: The sorting feature shall not expose additional sensitive information.

SEC-004: The feature shall not introduce unsafe HTML rendering.

8. Acceptance Criteria

The Change Request is successfully implemented when:

- **AC-001:** Administrator can access the sorting controls on the Admin Dashboard.
- **AC-002:** Administrator can sort student records in Ascending order.
- **AC-003:** Administrator can sort student records in Descending order.
- **AC-004:** Registration numbers are correctly ordered from lowest to highest and highest to lowest.
- **AC-005:** Student names are correctly ordered alphabetically in Ascending and Descending order.
- **AC-006:** Sorting is performed without unnecessary API requests.
- **AC-007:** The student table correctly reflects the selected sorting order.
- **AC-008:** Edit continues to affect the correct student.
- **AC-009:** Delete continues to affect the correct student.
- **AC-010:** Existing administrator authentication and authorization continue to work.

9. Expected Outcome

After implementation, administrators will be able to quickly organize and locate student records using Ascending and Descending order.

The sorting feature will improve the usability of the Admin Dashboard while keeping the existing API, database, authentication, authorization, and CRUD functionality unchanged.

10. Implementation Report Requirement

After implementation, the Final Exam Software Change Implementation Report shall include:

10.1 Implementation Summary

Describe how Ascending and Descending sorting was added to the Admin Dashboard and how the sorting operates on the already-loaded student records.

10.2 Affected Files

List all files modified or added during implementation.

10.3 Important Technical/Security Changes

Document:

- Client-side sorting implementation.
- Sorting fields used.
- Ascending and Descending comparison logic.
- Preservation of `student_id`.
- Confirmation that no backend or database changes were required.
- Confirmation that authentication and authorization were not modified.

10.4 Test Cases and Results

Test Case	Test Description	Expected Result	Status
TC-001	Login as Administrator	Admin Dashboard opens successfully	Pass/Fail
TC-002	Select Ascending order	Records are sorted correctly in Ascending order	Pass/Fail
TC-003	Select Descending order	Records are sorted correctly in Descending order	Pass/Fail
TC-004	Sort by Registration Number Ascending	Registration numbers appear lowest to highest	Pass/Fail
TC-005	Sort by Registration Number Descending	Registration numbers appear highest to lowest	Pass/Fail
TC-006	Sort by Student Name Ascending	Names appear alphabetically A–Z	Pass/Fail
TC-007	Sort by Student Name Descending	Names appear alphabetically Z–A	Pass/Fail
TC-008	Edit after sorting	Correct student is edited using <code>student_id</code>	Pass/Fail
TC-009	Delete after sorting	Correct student is deleted using <code>student_id</code>	Pass/Fail
TC-010	Authentication/Authorization	Existing access control continues to work	Pass/Fail

10.5 Acceptance Criteria Verification

Each acceptance criterion shall be verified and recorded as **Pass/Fail** with appropriate screenshots or evidence.

10.6 Relevant Screenshots / Evidence

The implementation report should include screenshots showing:

1. Admin Dashboard with student list.
2. Ascending order selected.
3. Student records displayed in Ascending order.
4. Descending order selected.
5. Student records displayed in Descending order.
6. Edit/Delete functionality after sorting.

10.7 Final Status

Status: Implemented / Tested / Accepted

10.8 Outstanding Issues

Document any known issues related to the sorting feature.

If no issues remain:

Outstanding Issues: None